

DEEPPFAKE DETECTION

ABSTRACT

This project involves the development of a deep learning model to detect deepfakes in video and image content. By leveraging advanced convolutional neural networks (CNNs) and deep learning techniques, the system aims to identify manipulated media with high accuracy, ensuring the authenticity of digital content.

TECHNOLOGIES USED

- **Machine Learning Model:** XceptionNet for robust feature extraction and classification .
- **Data Augmentation:** Techniques like GAN-based (Generative Adversarial Network) data augmentation to improve model robustness .
- **Video Frame Analysis:** Utilize OpenCV and TensorFlow to analyze frame-by-frame content in videos, enhancing detection precision.
- **Post-Processing:** Apply filtering and smoothing techniques to reduce false positives and improve overall accuracy.
- **Framework:** TensorFlow/Keras for model development .

APPLICATIONS USED

- **Social Media Platforms:** Enhancing content moderation by automatically detecting and flagging deepfake videos and images.
- **News Media:** Ensuring the authenticity of video and image content before publishing to prevent the spread of misinformation.
- **Legal and Forensics:** Providing tools for forensic analysis to verify the authenticity of digital evidence.